Contents

1 Fundamentals of Semiconductor Devices

Reference basis: NPTEL course - Fundamentals of Semiconductor Devices By Prof. Digbijoy N. Nath | IISc Bangalore

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1.1 Introduction to semiconductors

- Semiconductors are materials whose electrical conductivity lies between conductors and insulators and can be controlled using doping, temperature, and electric fields.
- They form the foundation of modern electronic devices such as diodes, transistors, and integrated circuits.
- Conduction occurs via two types of charge carriers: **electrons (negative)** and **holes (positive)**.
- Silicon (Si) is the most widely used semiconductor due to its stability and ease of fabrication.
- Their behavior is governed by quantum mechanics and band theory.

Moore's Law states that the number of transistors on an integrated circuit doubles approximately every 18–24 months(~1.5-2 years), leading to exponential growth in computing power. - This scaling results in higher performance, lower cost per transistor, and increased device density. - However, physical and thermal limits are slowing down this trend in modern technologies.

1.2 Energy bands

- In crystals, atomic energy levels combine to form continuous bands due to interaction between atoms.
- Energy Bands are ranges of allowed electron energies formed due to atomic orbital overlap in a crystal lattice.
- Electrons occupy energy bands: **valence band (VB)** and **conduction band (CB)**.
- Valence Band: Electron rich bands, holds bound electrons
- Conduction Band: Electron deficient bands, allows free electron movement
- The **bandgap (E)** is the energy difference between VB and CB.
- Small bandgap allows electrons to move to CB and conduct electricity.
- Insulators have large bandgap, while conductors have overlapping bands. Semiconductors have Moderate & achievable bandgap.
- Band structure determines electrical and optical properties.
- Holes: Absence of electrons
- Fermi level: Shows probability of finding electronics along the energy band gap

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1.3 Types of Semiconductors

- **Intrinsic Semiconductors**: Pure materials with equal electron and hole concentration ($ n = p = n_i $).
- Intrinsic semiconductor (extremely pure): Materials which have exactly same holes and electrons or Fermi level sits exactly in the middle
- **Extrinsic Semiconductors**: Doped materials with controlled carrier concentration.

1.3.1 Doping

- **Doping** is a process of adding impurities (either holes of electrons) to the intrinsic semiconductor to make it either n-type or p-type semiconductor
- This process is inherent only to the semiconductors. This cannot be done to conductors or insulators.
  - Eg. Adding Boron or Phosphorus can be added to the Si lattice
- Doping introduces donors (n-type) or acceptors (p-type). Based on these, we get 2 types of Extrinsic Semiconductors:
  - **n-type semiconductor**: Materials which have more electrons than holes or Fermi level is in the upper half
  - electrons are majority carriers in n-type semiconductors.
  - **p-type semiconductor**: Materials which have more holes than electrons or Fermi level is in the lower half
  - holes are majority carriers in p-type semiconductors.
- Doping enables control over conductivity.
- Doping enables control of electrical properties.
- Small doping significantly increases conductivity.
- Intrinsic carrier concentration depends on temperature and bandgap.

1.4 Band Structure

1.4.1 Band structure & fermi–dirac distribution

- Fermi–Dirac distribution gives the probability of an energy state being occupied:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- At $ T = 0K $, all states below $ E_F $ are filled and above are empty (step-like distribution).
- At finite temperature, electrons get thermally excited to higher energy states, smoothing the distribution.
- Fermi level shifts with doping: toward conduction band (n-type) and toward valence band (p-type).
- It directly determines electron and hole concentrations in semiconductors.

1.4.2 Density of states (DOS)

- Density of states (DOS) represents the number of available energy states per unit energy per unit volume.
- In 3D semiconductors, DOS increases as

$$\sqrt{E - E_c}$$

  above the conduction band edge.
- Carrier concentration is obtained by combining DOS with the Fermi–Dirac distribution.
- DOS depends on effective mass and dimensionality of the system.
- It is essential for calculating electron and hole populations in materials.

1.4.3 Equilibrium carrier concentration

- At thermal equilibrium, carrier concentrations follow:

$$np = n_i^2$$

- Charge neutrality ensures total positive and negative charges are balanced.
- Majority carriers are approximately equal to dopant concentration.
- Minority carriers are derived from intrinsic relations.
- Fermi level position uniquely determines equilibrium carrier densities.

1.4.4 Temperature dependence

- Intrinsic carrier concentration increases exponentially with temperature:

$$n_i \propto e^{-E_g/2kT}$$

- Three regions: freeze-out (low T), extrinsic (moderate T), intrinsic (high T).
- At high temperatures, intrinsic carriers dominate over dopants.
- Mobility decreases with temperature due to increased phonon scattering.
- Temperature strongly impacts device behavior and reliability.

1.4.5 High doping & incomplete ionization

- Heavy doping leads to **degenerate semiconductors**, where Fermi level enters a band.
- Bandgap narrowing occurs due to interaction between closely spaced dopants.
- At low temperatures, dopants may not fully ionize (incomplete ionization).
- Carrier concentration deviates from simple doping assumptions.
- Critical in heavily doped regions like MOSFET source/drain.

1.4.6 Carrier scattering & mobility

- Mobility defines how easily carriers move under an electric field:

$$\mu = \frac{q\tau}{m^*}$$

- Scattering mechanisms include phonon (lattice) scattering and impurity scattering.
- Higher temperature increases phonon scattering, reducing mobility.
- Higher doping increases impurity scattering, also reducing mobility.
- Mobility directly affects conductivity and switching speed of devices.

1.4.7 Low & high field transport

- At low electric fields, carrier velocity is proportional to electric field (Ohm's law).
- At high fields, velocity saturates due to increased scattering.
- Velocity saturation limits current in short-channel devices.
- High-field effects cause deviation from linear transport models.
- Important for modern nanoscale transistor design.

1.4.8 Drift–diffusion & traps

- Drift current arises due to electric field:

$$J_{drift} = qn\mu E$$

- Diffusion current arises due to carrier concentration gradient:

$$J_{diff} = qD\frac{dn}{dx}$$

- Einstein relation links diffusion and mobility:

$$\frac{D}{\mu} = \frac{kT}{q}$$

- Trap states capture carriers and influence recombination and leakage.
- Traps introduce non-ideal behavior such as reduced carrier lifetime.

1.4.9 Charge transport

- Total current is the sum of drift and diffusion currents.
- Drift dominates under strong electric fields; diffusion dominates under concentration gradients.
- Both electrons and holes contribute to current flow.
- Carrier mobility determines how quickly carriers respond to applied fields.
- These mechanisms form the basis of all semiconductor device operation.

1.4.10 Continuity equation

- Ensures conservation of charge:

$$\frac{\partial n}{\partial t} = G - R + \frac{1}{q}\nabla J$$

- Accounts for generation (G), recombination (R), and current flow (J).
- Used for both steady-state and transient analysis of devices.
- Fundamental equation in semiconductor device modeling and simulation.
- Connects microscopic carrier dynamics to measurable electrical current.

1.5 PN Junction

- A PN junction is formed by joining **p-type (hole-rich)** and **n-type (electron-rich)** semiconductor regions.
- Due to carrier concentration difference, electrons (from n-side) and holes (from p-side) **diffuse across the junction**.
- This diffusion leaves behind fixed ionized charges, forming a **depletion region** with no mobile carriers.
- An internal **electric field (built-in potential)** is created, which opposes further diffusion.
- At equilibrium, drift and diffusion currents balance, resulting in **zero net current**.
- The junction has a field and prevents the current from flowing with no field applied across the semiconductors.

1.5.1 Biasing of PN Junction

1. Forward Bias

- p-side connected to positive terminal, n-side to negative terminal.
- Reduces barrier potential and narrows depletion region.
- Allows majority carriers to cross the junction easily.
- Results in **large exponential current flow**:

$$I = I_s(e^{V/V_T} - 1)$$

- Turn-on voltage (~0.7V for Si) is required to significantly conduct.

2. Reverse Bias

- p-side connected to negative terminal, n-side to positive terminal.
- Increases barrier potential and widens depletion region.
- Prevents majority carrier flow.
- Only small **reverse saturation current** flows due to minority carriers.
- Acts like an insulator under normal conditions.

1.5.2 Depletion Region & Built-in Potential

- The depletion region acts as a **barrier** preventing free carrier movement.

- Built-in potential $ V_{bi} $ depends on doping levels:

$$V_{bi} \propto \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

- Width of depletion region changes with applied voltage.
- This region is crucial for controlling current flow in the device.
- Acts like an insulating layer between p and n regions.
- When voltage is applied, it takes some voltage to overcome the depletion region before current starts flowing. This is called as turn on voltage.
- The semiconductor becomes operational when more voltage is applied than turn on voltage.

1.5.3 Breakdown Mechanisms

- At high reverse voltage, junction undergoes **breakdown**, causing large current.

1.5.4 Types:

- **Zener (Tunneling) Breakdown**
  - Occurs in heavily doped junctions at low voltage.
  - Strong electric field enables quantum tunneling.
- **Avalanche Breakdown**
  - Occurs in lightly doped junctions at high voltage.
  - Carriers gain energy and generate more carriers via collisions.
- Breakdown can damage the device unless properly controlled (e.g., Zener diodes).

1.6 Applications of PN Junction

1.6.1 Rectifier (Diode)

- Converts AC to DC by allowing current in one direction only.
- When connected in Forward Bias, the junction allows the current to flow.
- In Reverse Bias, prevents current flow except for a small leakage current.
- Used in power supplies and electronic circuits.

1.6.2 LED (Light Emitting Diode)

- An LED (Light Emitting Diode) is a PN junction diode that emits light when forward biased.
- Electrons and holes recombine at the junction, releasing energy as photons (light).
- Emits light when forward biased due to **radiative recombination**.
- Electron-hole recombination releases photons:

$$E_{photon} \approx E_g$$

- Color depends on semiconductor bandgap.
- Used in displays, lighting, and indicators.

1.6.3 Photodetector (Photodiode)

- A photodetector is a PN junction device that converts light into electrical current.
- When light photons strike the junction, they generate electron-hole pairs in the depletion region.
- Under reverse bias, these carriers are quickly swept across the junction, producing photocurrent.
- Also can be used in any spectrum: IR, UV, Infrared etc.
- Used in optical communication and sensors.
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1.6.4 Solar Cell

- A solar cell is a PN junction device that converts sunlight directly into electrical energy using photovoltaic effect.
- Photons striking the junction create electron-hole pairs, generating current.
- It operates under illumination without external bias, using built-in potential to separate charges.
- Only sunlight is the input spectra.
- Used in renewable energy systems.

1.7 Schottky Junction (Metal–Semiconductor Junction)

- A Schottky junction is formed when a **metal is brought into contact with a semiconductor** (typically n-type).
- Instead of a p–n interface, it creates a **metal–semiconductor interface** with unique electrical behavior.
- Charge transfer occurs at the interface, leading to formation of a **Schottky barrier**.
- It is also called a **Schottky diode** when used as a device.

1.7.1 Formation & Barrier

- When metal and semiconductor come in contact, electrons flow to align their Fermi levels.
- This creates a **depletion region in the semiconductor side only** (not in the metal).
- A potential barrier called the **Schottky barrier height ($ _B $)** is formed.
- Barrier height depends on **metal work function and semiconductor properties**.
- The built-in electric field prevents further carrier diffusion at equilibrium.

1.7.2 Operation

1. Forward Bias

- Reduces the barrier height at the interface.
- Majority carriers (electrons in n-type) easily cross into the metal.

- Results in **low forward voltage drop (~0.2–0.3 V)** compared to silicon PN diodes (~0.7 V).
- Current increases rapidly with applied voltage.

2. Reverse Bias

- Increases barrier height and widens depletion region.
- Prevents majority carrier flow.
- Only small leakage current flows (higher than PN junction).
- No significant charge storage occurs.

1.7.3 Key Characteristics

- **Majority carrier device** → no minority carrier storage.
- **Fast switching speed** due to absence of charge storage.
- **Low forward voltage drop** → lower conduction losses.
- **Higher reverse leakage current** compared to PN junction.
- Barrier properties depend on **choice of metal**.

1.7.4 Advantages

- High-speed operation (ideal for RF and switching circuits).
- Low power loss due to low forward voltage.
- Simple structure compared to PN junction.

1.7.5 Limitations

- Higher reverse leakage current.
- Lower breakdown voltage compared to PN diodes.
- Sensitive to temperature variations.

1.8 Transistors

- A transistor is a **three-terminal semiconductor device** used to control current flow and signal behavior.
- It operates by using a **small input signal** to control a much larger current between two terminals.
- The three terminals depend on type of transistor:
  - BJT: Emitter, Base, Collector
  - FET: Source, Gate, Drain
- Transistors are the **fundamental building blocks** of modern electronic circuits.

1.8.1 Key Functions / Applications

- **Switch**: Turns current ON/OFF (used in digital circuits).
- **Amplifier**: Increases signal strength (analog circuits).
- **Digital Logic**: Forms logic gates and processors (CMOS technology).
- **Sensors**: Used in sensing circuits and signal conditioning.
- **Memory**: Stores data in SRAM, DRAM, and Flash technologies.

1.8.2 Types of Transistors

1. Field Effect Transistor (FET)

- A **voltage-controlled device** where current is controlled by an electric field at the gate.
- Terminals: **Source, Gate, Drain**.
- Gate voltage controls the conductivity of the channel between source and drain.
- Very high input impedance (almost no gate current).
- Examples: **MOSFET, JFET** → widely used in digital electronics.

2. Bipolar Junction Transistor (BJT)

- A **current-controlled device** that uses both electrons and holes (bipolar conduction).
- Terminals: **Emitter, Base, Collector**.
- A small base current controls a large collector current (current amplification).
- Types: **NPN and PNP transistors**.
- Used in amplification and analog circuits.

1.9 Bipolar Junction Transistor

- A BJT (Bipolar Junction Transistor) is a three-terminal semiconductor device (emitter, base, collector) that can amplify or switch signals.
- It uses both electrons and holes as charge carriers, hence the term "bipolar".
- **Base (B)** Thin, lightly doped region that controls carrier flow from emitter to collector by regulating recombination.
- **Emitter (E)** Heavily doped region that injects majority carriers into the base.
- **Collector (C)** Moderately doped region that collects carriers from the base and delivers output current.
- **Currents from E, B, C** : Emitter current splits into base and collector currents: $ I_E = I_B + I_C $
- **Gain (Current Gain)** : Ability of BJT to amplify current, defined as ratio of collector current to base current:
$$\beta = \frac{I_C}{I_B}$$
- **Beta ( )** : Common-emitter current gain representing amplification capability of the transistor:
$$\beta = \frac{I_C}{I_B}$$
- **Base Transport Factor ( )** : Fraction of carriers injected from emitter that successfully reach the collector through the base:
$$\alpha_T = \frac{I_C}{I_E'}$$

- **Emitter Injection Efficiency ( )** : Fraction of emitter current that contributes to useful carrier injection:

$$\gamma = \frac{\text{electron injection}}{\text{total emitter current}}$$

- **Transport Ratio / Common-base Gain ( )** : Fraction of emitter current that becomes collector current:

$$\alpha = \frac{I_C}{I_E}$$

- **Important Relation** :

$$\beta = \frac{\alpha}{1 - \alpha}$$

1.9.1 BJT Operation

- When a small positive voltage is applied to the base relative to the - emitter, electrons are injected from the emitter into the base.
- These electrons cross the thin base region and are collected by the - collector, creating a large collector current.
- Thus, a small base current controls a much larger collector-emitter current, enabling amplification or switching.
- Emmiter-Base junction is placed in forward bias & Base-Collector is placed in reverse bias.
- A small current at the base controls a much larger current between collector and emitter.
- For best BJT operation & improving gain, Beta & other parameters, base should be narrow & lightly doped.
- BJTs has 2 types:
  1. NPN: An NPN transistor has an n-type emitter, p-type base, and n-type - collector.
  2. PNP: An PNP transistor has an p-type emitter, n-type base, and p-type - collector.

1.10 MOSFET

1.10.1 MOS (Metal-oxide-semiconductor) CAPACITOR

- A MOS capacitor consists of a Metal–Oxide–Semiconductor structure where the oxide acts as an insulator and the gate voltage controls charge at the semiconductor surface.
- Semiconductor can be p-type or n-type doped.
- Applying a gate voltage changes the surface potential, causing redistribution of carriers near the oxide-semiconductor interface.
- Depending on gate voltage polarity and magnitude, the device operates in accumulation, depletion, or inversion.
- No current flows through the oxide (ideal case); only charge rearrangement occurs.
- MOS capacitor behavior is fundamental to understanding MOSFET operation.

- Accumulation: Occurs when gate voltage attracts majority carriers to the surface, increasing carrier concentration near the interface.
- Depletion: Occurs when gate voltage repels majority carriers, leaving behind fixed ionized charges and forming a depletion region.
- Inversion: Occurs when sufficient gate voltage attracts minority carriers, forming a conductive channel at the surface.
- Threshold Voltage V_T: The gate voltage at which strong inversion begins and a conducting channel is formed.

1.10.2 MOSFET definition

- A MOSFET (Metal–Oxide–Semiconductor Field-Effect Transistor) is a three-terminal device (Gate, Source, Drain) where current between source and drain is controlled by gate voltage.
- It is derived from the MOS capacitor by adding two heavily doped regions (source and drain) in the semiconductor.
- The gate voltage controls the formation of a conductive channel at the semiconductor surface.
- It is a voltage-controlled device with very high input impedance due to the insulating oxide.
- MOSFETs are the fundamental building blocks of modern digital and analog circuits.

1.10.3 Relation to MOS Capacitor

- The MOSFET structure is essentially a MOS capacitor with source and drain added for current flow.
- The same surface charge control principles (accumulation, depletion, inversion) apply.
- Channel formation in MOSFET occurs under **inversion condition** of MOS capacitor.
- Threshold voltage of MOS capacitor directly determines when MOSFET turns ON.
- Thus, MOSFET operation is an extension of MOS capacitor electrostatics.

1.10.4 Operation of MOSFET (nMOS example)

1. Cutoff Region (OFF)

- When $ V_{GS} < V_T $, no inversion layer forms and no conduction path exists.
- Only very small leakage current flows.

2. Linear / Triode Region

- When $ V_{GS} > V_T $ and $ V_{DS} $ is small, a continuous channel forms.
- Device behaves like a voltage-controlled resistor.
- Drain current:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

3. Saturation Region

- When $ V_{DS} (V_{GS} - V_T) $, channel pinches off near drain.
- Current becomes almost independent of $ V_{DS} $.
- Drain current:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

1.10.5 Key Physical Insight

- Gate voltage controls surface potential → creates inversion layer → forms channel → enables current flow.
- MOSFET current is controlled purely by electric field (no gate current ideally).
- Device transitions from **no channel → resistive channel → pinch-off region** as voltages change.

1.10.6 Important Parameters

- $ V_T $: Threshold voltage (onset of inversion)
- $$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

  : Oxide capacitance per unit area
- $ _n $: Carrier mobility
- $ W/L $: Device geometry (width/length ratio)
- Gate voltage controls channel formation.
- Regions: cutoff, linear, saturation.
- Current depends on gate voltage.
- Short-channel effects degrade performance.
- Most important device in digital electronics.

1.11 Compound Semiconductors

- Compound semiconductors are materials formed by combining two or more elements (typically from III–V or II–VI groups) such as GaAs, InP, and GaN.
- They offer superior electronic and optical properties compared to silicon, especially for high-speed and optoelectronic applications.
- Many compound semiconductors have a **direct bandgap**, enabling efficient light emission and absorption.
- Their properties can be engineered by changing composition (e.g., AlGaAs, InGaN).
- Widely used in LEDs, lasers, RF devices, and high-frequency electronics.
- Compound semiconductors enable **high-speed + optical + high-power applications** where silicon falls short.

Group / Material Type	Example Materials	Key Properties	Typical Applications
III–V Arsenides	GaAs, AlGaAs	Direct bandgap, high electron mobility, moderate bandgap (~1.4 eV)	RF devices, microwave circuits, LEDs, laser diodes

Group / Material Type	Example Materials	Key Properties	Typical Applications
III–V Phosphides	InP, GaP	High electron velocity, good thermal stability	High-speed electronics, optical communication, lasers
III–V Nitrides	GaN, AlGaN, InGaN	Wide bandgap, high breakdown field, high temperature stability	Power electronics, LEDs (blue/white), RF amplifiers
III–V Antimonides	InSb, GaSb	Very narrow bandgap, very high mobility	Infrared detectors, thermal imaging, sensors
II–VI Semiconductors	CdS, CdSe, ZnSe	Direct bandgap, strong optical interaction	Photodetectors, LEDs, display technologies
Ternary Compounds	AlGaAs, InGaN	Tunable bandgap via composition	LEDs, laser diodes, heterostructure devices
Quaternary Compounds	InGaAsP, AlGaInP	Precise bandgap engineering, lattice matching	Fiber-optic communication, high-efficiency lasers
Heterostructures	GaAs/AlGaAs, InP/InGaAs	Band offset engineering, carrier confinement	HEMTs, HBTs, high-speed transistors
Wide Bandgap Semiconductors	GaN, SiC	High breakdown voltage, high temperature operation	Power devices, EV electronics, high-frequency systems
Narrow Bandgap Semiconductors	InSb, HgCdTe	Sensitive to IR radiation, low energy gap	Infrared photodetectors, night vision systems

- **Direct bandgap → optoelectronics (LEDs, lasers)**
- **High mobility → high-speed RF devices**
- **Wide bandgap → power electronics**
- **Narrow bandgap → infrared detection**

1.11.1 Key Properties

- **High electron mobility** → faster carrier transport and high-speed device operation.
- **Direct bandgap (in many materials)** → efficient light emission (used in LEDs and lasers).
- **Wide bandgap (e.g., GaN)** → high power and high-temperature operation.
- **Bandgap engineering** → allows tuning of optical and electrical properties.
- **Heterostructure compatibility** → enables advanced device design.

1.11.2 Comparison with Silicon

- Compound semiconductors have higher mobility and better optical properties than silicon.

- Silicon has indirect bandgap → poor light emission, while many compounds are direct bandgap.
- Compound semiconductors are more expensive and harder to fabricate.
- Silicon dominates digital ICs, while compound semiconductors dominate optoelectronics and RF.
- Trade-off exists between cost, performance, and manufacturability.

1.11.3 Applications

- **Optoelectronics**: LEDs, laser diodes (GaAs, InGaN).
- **High-frequency devices**: RF amplifiers, microwave circuits (GaAs, InP).
- **Power electronics**: GaN-based high-voltage devices.
- **Solar cells**: High-efficiency multi-junction cells.
- **Sensors & photodetectors**: Fast optical detection systems.

1.12 Solar Cells

- A solar cell is a semiconductor device that converts solar light energy into electrical energy using the photovoltaic effect.
- It is typically a **p–n junction device**, where incident photons generate electron–hole pairs.
- The built-in electric field in the depletion region separates these carriers, producing current.
- When connected to a load, this current flows as usable electrical power.
- Solar cells are the fundamental units of solar panels used in renewable energy systems.
- **Solar cells do not require external voltage application for generating current.**
- **Short-Circuit Current (Isc)** : The current flowing through the solar cell when the output terminals are shorted (i.e., $ V = 0 $), representing the maximum current generated under illumination.
- **Open-Circuit Voltage (Voc)** : The voltage across the solar cell when no external current flows (i.e., $ I = 0 $), representing the maximum voltage developed due to carrier separation.

1.12.1 Working Principle

- Photons with energy greater than bandgap ($ h E_g $) generate electron–hole pairs.
- The internal electric field drives electrons toward the n-side and holes toward the p-side.
- This separation prevents recombination and creates a voltage across the junction.
- External circuit allows current to flow, delivering power.
- Continuous illumination maintains carrier generation and output.

1.12.2 Loss Mechanisms

- Recombination of carriers before collection reduces efficiency.
- Reflection losses prevent light from entering the device.
- Resistive losses in contacts and material reduce output power.
- Thermalization losses occur when excess photon energy is lost as heat.

- Imperfect absorption limits carrier generation.

1.12.3 Types of Solar Cells

- **Silicon Solar Cells**: Most common, reliable, moderate efficiency.
- **Thin-Film Solar Cells**: Lightweight, flexible, lower efficiency.
- **Compound Semiconductor Cells**: High efficiency (GaAs, multi-junction).
- **Perovskite Solar Cells**: Emerging, high efficiency potential.

1.13 Photodetectors

- A photodetector is a semiconductor device that converts incident light with a certain wavelength (Infrared, visible, Ultraviolet etc.) into an electrical signal by generating electron–hole pairs.
- It typically operates using a **p–n or p–i–n junction**, where light absorption creates carriers.
- The built-in or applied electric field separates these carriers, producing current.
- Output signal is proportional to incident optical power.
- Widely used in optical communication, sensing, and imaging systems.

1.13.1 Working Principle

- Incident photons with energy $ h E_g $ generate electron–hole pairs.
- These carriers are separated by the electric field in the depletion region.
- Electrons move toward n-side and holes toward p-side, creating photocurrent.
- Reverse bias is often applied to increase depletion width and speed.
- Faster response is achieved with thinner active regions and strong fields.

1.13.2 Types of Photodetectors

- **PN Photodiode**: Basic structure, moderate speed and sensitivity.
- **PIN Photodiode**: Wider depletion region give better sensitivity and faster response.
- **Avalanche Photodiode (APD)**: Internal gain via impact ionization → high sensitivity.
- **Photoconductor**: Conductivity changes with light, slower but simple.

1.13.3 Key Performance Parameters

- **Responsivity (R)**: Output current per unit optical power:

$$R = \frac{I_{ph}}{P_{opt}}$$

- **Quantum Efficiency ( )**: Fraction of photons converted into carriers.
- **Dark Current**: Leakage current in absence of light.
- **Response Time / Bandwidth**: Speed of detector operation.
- **Noise**: Limits sensitivity and minimum detectable signal.

1.13.4 Important Concepts

- Higher absorption coefficient results in more carrier generation.
- Larger depletion region yield in higher efficiency but slower response (trade-off).
- Reverse bias improves speed but increases noise.
- Avalanche multiplication enhances signal but adds excess noise.
- Material bandgap determines wavelength sensitivity.

1.13.4.1 Photodetectors vs Solar Cells

Feature	Photodetectors	Solar Cells
Purpose	Detect light and convert it into an electrical signal	Convert light into usable electrical power
Primary Goal	Sensitivity and speed	Maximum energy conversion efficiency
Operation Mode	Usually operated in **reverse bias** for fast response	Operated in **zero bias or forward bias** (power generation mode)
Output	Small signal current proportional to light intensity	Continuous power output (voltage + current)
Speed	Very fast (ns–ps range for high-speed devices)	Relatively slow (not optimized for speed)
Responsivity	High sensitivity to small optical signals	Designed for maximum total energy capture
Efficiency Focus	Detection efficiency (quantum efficiency)	Power conversion efficiency ( )
Structure	PN, PIN, Avalanche photodiodes	Mainly PN junction (sometimes multi-junction)
Gain Mechanism	Can include internal gain (APD via avalanche)	No internal gain mechanism
Noise Consideration	Noise is critical (limits detection capability)	Less critical compared to power output
Applications	Optical communication, sensors, imaging	Solar panels, renewable energy systems

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1.14 Recombination in LEDs

- Recombination is the process where electrons from the conduction band combine with holes in the valence band, releasing energy in the form of photons (light).
- This is primarily **radiative recombination**, which is the desired mechanism for light emission.
- Occurs in the active region (usually a direct bandgap semiconductor).
- Efficiency of LED depends on how many recombinations produce photons.
- Direct bandgap materials (e.g., GaN, GaAs) are used to maximize light emission.

1.14.1 Types of Recombination

- **Radiative Recombination**: Electron-hole recombination that emits a photon (useful for LEDs).
- **Non-radiative (SRH) Recombination**: Energy lost as heat via trap states, reducing efficiency.
- **Auger Recombination**: Energy transferred to another carrier instead of emitting light, dominant at high carrier densities.

1.14.2 Key Concepts

- Radiative recombination rate is proportional to electron and hole concentrations:

$$R_{rad} \propto np$$

- Internal Quantum Efficiency (IQE):

$$IQE = \frac{\text{radiative recombination}}{\text{total recombination}}$$

- Higher radiative recombination results in brighter LED output.
- Non-radiative processes reduce efficiency and generate heat.
- Carrier confinement (using heterostructures) improves recombination efficiency.
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1.15 LED (Light Emitting Diode)

- An LED is a semiconductor **p–n junction device** that emits light when forward biased due to electron–hole recombination.
- It uses **direct bandgap materials** (e.g., GaN, GaAs) to efficiently convert electrical energy into light.
- The emitted light wavelength (color) depends on the bandgap of the material.
- LEDs are widely used in displays, lighting, and indicators due to high efficiency and long life.
- It is an optoelectronic device combining electrical and optical phenomena.

1.15.1 Operation Principle

- When forward bias is applied, electrons from n-side and holes from p-side are injected into the junction.
- These carriers recombine in the active region.
- In direct bandgap materials, recombination releases energy as photons (light emission).
- Photon energy is approximately equal to bandgap:

$$E_{photon} \approx E_g = h\nu$$

- Continuous carrier injection results in continuous light output.

1.15.2 Key Concepts

- **Forward Bias** is essential for carrier injection and light emission.
- **Radiative recombination** produces light, while non-radiative processes reduce efficiency.
- **Internal Quantum Efficiency (IQE)** measures how efficiently carriers produce photons.
- **External Quantum Efficiency (EQE)** also includes light extraction efficiency.
- Device design focuses on maximizing light output and minimizing losses.

1.15.3 Important Factors Affecting LED Performance

- Material choice (direct bandgap required).
- Carrier recombination efficiency.
- Light extraction efficiency (internal reflections reduce output).
- Temperature (higher temperature reduces efficiency).
- Device structure (heterostructures improve confinement).

1.16 Advanced transistors

1.16.1 Transistors for Power Electronics

- Power transistors are semiconductor devices designed to handle **high voltage, high current, and high power levels** efficiently.
- They are used as switches or amplifiers in power conversion systems.
- Unlike small-signal devices, they are optimized for **low losses and high reliability**.
- Common types include **Power MOSFETs, IGBTs, and BJTs**.
- Widely used in power supplies, inverters, motor drives, and EV systems.

1.16.1.1 Key Requirements

- **High breakdown voltage** to withstand large voltages in OFF state.
- **Low ON-state resistance** to reduce conduction losses.
- **High current handling capability** for power applications.
- **Efficient heat dissipation** to manage thermal effects.
- **Fast switching capability** to minimize switching losses.

1.16.1.2 Operation Principle

- Power transistors operate mainly as **switches** (ON/OFF states).
- In ON state low resistance path allows current flow.
- In OFF state high resistance blocks current.
- Switching between states enables energy control and conversion.
- Efficiency depends on minimizing conduction and switching losses.

1.16.1.3 Types of Power Transistors

- **Power MOSFET**: Fast switching, low gate power, used in low–medium voltage.
- **IGBT (Insulated Gate Bipolar Transistor)**: Combines MOSFET control with BJT conduction, used in high power.
- **Power BJT**: High current capability but slower switching, less commonly used now.

1.16.1.4 Loss Mechanisms

- **Conduction Loss**: Due to ON resistance or voltage drop.
- **Switching Loss**: During transitions between ON and OFF states.
- **Leakage Loss**: Small current in OFF state.
- **Thermal Loss**: Heat generated affects efficiency and reliability.

1.16.2 Transistors for Memory

1.16.2.1 What are Memory Transistors?

- Memory transistors are semiconductor devices designed to **store data (bits) by retaining charge or state**.
- They form the basic building blocks of memory technologies like **SRAM, DRAM, and Flash**.
- Unlike logic transistors, they are optimized for **data retention, stability, and low leakage**.
- Information is stored as either charge, voltage level, or stored state.
- Used in caches, RAM, and non-volatile storage systems.

1.16.2.2 Operation Principle

- Memory operation involves **write, store, and read** cycles.
- During write, charge is stored in a node or capacitor.
- During hold, the device retains stored information for a certain duration.
- During read, stored data is sensed without disturbing it significantly.
- Data is represented as binary states (0 or 1).

1.16.2.3 Types of Memory Transistor Structures

- **SRAM (Static RAM)**: Uses multiple transistors (typically 6T) to store a stable state using feedback.
- **DRAM (Dynamic RAM)**: Uses one transistor + capacitor; requires periodic refresh.
- **Flash Memory**: Uses floating gate MOSFET to store charge permanently (non-volatile).
- **EEPROM**: Allows electrical erasing and reprogramming.

1.17 Microelectronic Fabrication

- Microelectronic fabrication is the process of creating semiconductor devices and integrated circuits on a wafer (typically silicon).
- It involves a sequence of physical and chemical processes to build structures at micro/nano scale.
- Devices are fabricated layer-by-layer using precise patterning techniques.
- Fabrication is carried out in cleanroom environments to avoid contamination.
- It is the foundation of modern VLSI and semiconductor manufacturing.
- Fabrication is a **repetitive cycle of patterning + modification + deposition**.
- Precision at nanometer scale is critical for device performance.
- Alignment between layers is crucial for circuit functionality.
- Process variations can significantly impact device characteristics.
- Yield (percentage of working chips) is a key manufacturing metric.

1.17.1 Key Process Steps

- **Oxidation**: Growth of silicon dioxide (SiO) layer for insulation and protection.
- **Photolithography**: Pattern transfer using light, masks, and photoresist.
- **Etching**: Removal of material (wet or dry) to define structures.
- **Doping (Diffusion / Ion Implantation)**: Introduction of impurities to form p-type or n-type regions.
- **Deposition**: Addition of material layers (chemical or physical).

1.17.2 Supporting Processes

- **Annealing**: Heat treatment to repair damage and activate dopants.
- **Planarization (CMP)**: Smoothening wafer surface for uniform layers.
- **Metallization**: Formation of metal interconnects for electrical connections.
- **Passivation**: Protective layer to prevent contamination and damage.

1.17.3 Challenges

- Scaling down device size increases fabrication complexity.
- Contamination can destroy devices → strict cleanroom control required.
- Process variations affect performance and reliability.
- High cost of fabrication facilities (fabs).
- Thermal and material limits at nanoscale.

1.18 Books and references

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ii. Solid State Electronic Devices, by Ben Streetman and Sanjay Banerjee, Prentice Hall.
iii. Introduction to Semiconductor Materials and Devices, by M. S. Tyagi, Wiley Publications.
